

# Ammar Vora

[ammavora@mail.utoronto.ca](mailto:ammavora@mail.utoronto.ca) | [+15878924403](tel:+15878924403) | [linkedin.com/in/ammavora](https://linkedin.com/in/ammavora) | [github.com/Ammar-V](https://github.com/Ammar-V) | [ammavora.com](https://ammavora.com)

## SKILLS

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**Programming Languages:** Python, C/C++, Java, JavaScript, HTML/CSS, Assembly, Verilog HDL

**Frameworks & Libraries:** PyTorch, OpenCV, NumPy, ROS/ROS2, Node.js, React.js, Express.js, JavaFX

**Tools:** Git, GitHub, Gitlab, Linux, RViz, Gazebo, MongoDB, Google Firestore, Firebase, Google Cloud Platform

## EDUCATION

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**University of Toronto** (Toronto, ON, Canada)

September 2021 - April 2026

Bachelor's of Applied Science and Engineering – Computer Engineering with double minor in AI and Robotics

*Cumulative GPA:* 3.84/4.0 (Honors Student, Dean's List, Engineering Alumni Network Scholarship)

*Relevant Coursework:* Applied Fundamentals of Deep Learning, Robot Modelling and Control, Computer Architecture

*Activities:* President of the University of Toronto Robotics Association, Head Leader for the Orientation Committee

## EXPERIENCE

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**Tenstorrent Inc.** (Boston, Massachusetts)

May 2024 - Present

Machine Learning Engineer Intern 

- Contributing to an **open-source** low-level kernel library in C++ to support fast inference on custom AI accelerators
- Leveraged current research on overlapped communication and computation in **distributed systems** to implement a new operation that achieves **87%** overlap and leads to a **1.5x** reduction in kernel duration
- Optimized device runtime for an operation used in an LLM by **8x** using **kernel fusion** and **multi-core parallelism**
- Analyzed profiling data to optimize and reduce overhead for a **memory-bound** operation, achieving a **4x** speedup
- Developing Python scripts to automate testing and guard against any codebase regressions in CI/CD workflows

**Medical Computer Vision and Robotics Lab** (University of Toronto)

May 2023 - August 2023

Undergraduate Researcher

- Developing an image processing pipeline to generate 3D reconstructions of the Larynx to quantify a rare disease
- Fine-tuned hyperparameters while training a CNN based trachea detection model, resulting in **94%** test accuracy
- Conducted data collection and labeled endoscope images using the Computer Vision Annotation Tool (CVAT)

**AUToronto, Self-Driving Car Team** (GM-SAE Autodrive Challenge II)

September 2022 - June 2024

Member of Perception team 

- Collaborated on restructuring and testing a lane detection pipeline, ranking **1st** in the Perception Challenge
- Built a lane line clustering algorithm using OpenCV, leading to a **robust** and **real time** system running at **30 fps**
- Tuned a 3D LiDAR based object detection pipeline, decreasing false positives in railroad bar and sign detections

**Autonomous Rover Team** (Intelligent Ground Vehicles Competition)

September 2021 - August 2023

Lead of Computer Vision team 

- Spearheaded development of the object detection pipeline, resulting in **3rd** place in the design presentation
- Optimized an algorithm to project 2D object detections into 3D using C++, accelerating runtime by **3 times**
- Optimized architecture of the U-Net model in PyTorch, boosting inference speed by **10x** while retaining accuracy
- Designed an automatic pothole dataset generation pipeline in Python, **saving over 50 hours** in data labeling

## PROJECTS

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**Birds** - Applied Fundamentals in Deep Learning 

May 2023 - August 2023

- Trained a Generative Adversarial Network with CNNs in Pytorch to animate the flying motion of a bird in an image

**City Explorer** - Software Communication and Design 

January 2023 - April 2023

- Designed fast object-oriented APIs and algorithms in C++ that visualize OpenStreetMap data in **5 milliseconds**
- Multi-threaded a Genetic Algorithm for the traveling salesman problem and enhanced runtime speed by **400%**

**Rescue Ranger** - UofTHacks X (Awarded 1st Place) 

January 2023 - January 2023

- Engineered an autonomous rover and a **full-stack** web application to help with Search and Rescue missions